

Completeness Audit

A data-quality map of the WDI database

Which countries, which series, and which years have missing data — and how much.

1,510 Series analyzed	265 Countries / regions	66 Years covered (1960–2025)
347,097 Country–series pairs with a gap	1,397 / 1,510 Series with at least one gap	~113 Series 100% complete

`data-for-economists` project — audit produced with the R package `WDI`, via exhaustive batch download of the full database.

Methodology

The entire World Development Indicators database was downloaded via the World Bank API (R package `WDI`), in batches of 20 series to work around API instability on large volumes. For each indicator × country × year combination, a present/missing status was computed, then grouped by (country, series) pair to precisely identify missing year ranges.

⚠ **IMPORTANT METHODOLOGICAL NOTE**

Completeness rates in this report are calculated over the full 1960–2025 window. Many WDI series only started being collected several decades after 1960 (environmental, governance, or recent financial indicators). A rate of 50% therefore does **not** mean that half of the *relevant* data is missing — it mechanically reflects this extended time window. For illustration, even Peru, one of the best-covered countries, has only 77 out of 1,397 series that are completely empty; most series have partial rather than zero coverage, concentrated in recent decades.

Processing steps

- Download all ~1,510 series × all countries × 1960–2025, in batches of 20 series
- Reshape to long format: one row = indicator × country × year
- Flag each combination as present or missing
- Group missing years into readable ranges per (country, series) pair

Completeness by series

Out of 1,510 series, 1,397 have at least one missing value. The least complete concern very specific construction-permit indicators, almost never reported; the most complete concern basic demographic breakdowns collected almost systematically.

10 least complete series

1	IC.BRE.BE.OS — Building permits, other		0.3%
2	IC.BRE.BE.P1 — Building permits, procedure 1		0.3%
3	IC.BRE.BE.P2 — Building permits, procedure 2		0.3%
4	IC.BRE.BE.P3 — Building permits, procedure 3		0.3%
5	IC.BRE.BI.OS — Building permits (index), other		0.3%
6	IC.BRE.BI.P1 — Building permits (index), proc. 1		0.3%
7	IC.BRE.BI.P2 — Building permits (index), proc. 2		0.3%
8	IC.BRE.BI.P3 — Building permits (index), proc. 3		0.3%
9	IC.BRE.BL.OS — Building permits (list), other		0.3%
10	IC.BRE.BL.P1 — Building permits (list), proc. 1		0.3%

10 most complete series

1	SP.POP.65UP.MA.ZS — Population 65+, male (%)		99.6%
2	SP.POP.65UP.TO.ZS — Population 65+, total (%)		99.6%
3	SP.POP.7074.FE.5Y — Population female 70–74		99.6%
4	SP.POP.7074.MA.5Y — Population male 70–74		99.6%
5	SP.POP.7579.FE.5Y — Population female 75–79		99.6%

6	SP.POP.7579.MA.5Y — Population male 75–79		99.6%
7	SP.POP.80UP.FE.5Y — Population female 80+		99.6%
8	SP.POP.80UP.MA.5Y — Population male 80+		99.6%
9	SP.POP.TOTL.FE.ZS — Population, female (%)		99.6%
10	SP.RUR.TOTL.ZS — Rural population (%)		99.6%

Full detail for all 1,510 series: see [completeness_stats_by_series.csv](#)

Completeness by country

The least complete entities are very small territories or special classification cases. Among large countries, the maximum completeness observed plateaus around 50% (see methodological note) — Latin American and Southeast Asian countries dominate this ranking.

10 least complete countries / regions

1	Not classified		0%
2	St. Martin (French part)		6.3%
3	Channel Islands		8.6%
4	Isle of Man		8.6%
5	Sint Maarten (Dutch part)		9%
6	Northern Mariana Islands		9.2%
7	Gibraltar		10.3%
8	Curaçao		10.7%
9	American Samoa		11%
10	Liechtenstein		11.3%

10 most complete countries

1	Peru		49.8%
2	Mexico		49.5%
3	Colombia		49.3%
4	India		48%
5	Indonesia		47.8%
6	Ecuador		47.7%
7	Dominican Republic		47.6%
8	Brazil		47.6%
9	Egypt, Arab Rep.		47.5%
10	Thailand		47.3%

Full detail for all 265 countries/regions: see [completeness_stats_by_country.csv](#)

Completeness by year

Overall completeness improves over time: few series existed in the 1960s, and coverage broadened as new indicators were created and data collection expanded to more countries.

1960		9.4%
1961		10.9%
1962		11.2%
1963		12.1%
1964		12.6%

Full detail for all 66 years: see [completeness_stats_by_year.csv](#)

Going further

This document summarizes general trends. For a fine-grained exploration — exactly which years are missing for a given country and series — see `missing_values_report_WDI.csv`, which details, for each of the 347,097 affected country-series combinations, the exact list of years without data. The full R script behind this audit (`audit_WDI_final_en.R`) is also available in this folder to reproduce or extend this analysis.